Performance Testing of a Centrifugal Pump

Characteristic curves, efficiency, dimensionless performance, and specific speed at three rotational speeds

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Team	Group A4: Luis Ramirez, Macklin Horner, Zachary Klein, Larimel Rosario, John Wahba	**Equipment**	GUNT HM 150.04 centrifugal pump trainer on HM 150 base module, suction and delivery Bourdon gauges, volumetric tank, stopwatch

PROJECT SUMMARY

A single-stage radial centrifugal pump was tested at 1800, 2500, and 3000 rpm. At each speed the delivery valve was closed in steps from fully open to shut-off while suction and delivery pressure, collected volume, and motor power were recorded. From these readings the head, hydraulic power, brake power, and efficiency were found and plotted against capacity. At 3000 rpm the pump reached a shut-off head of 28.2 m and a best efficiency of 21% at 44 L/min and 16.5 m. The specific speed at the best-efficiency point ranged from $N_s \approx$ 513 to 766 (US units), placing the pump at the low end of the radial-flow range, which explains its modest efficiency.

1. Introduction

Centrifugal pumps are the most widely used pumps in engineering, moving water through supply and sewage systems, HVAC loops, boiler feed lines, irrigation, and process plants. Unlike a positive-displacement pump, the flow a centrifugal pump delivers depends on the head it must overcome, so its behavior is described by characteristic curves of head, power, and efficiency against capacity. The goal of this experiment was to measure those curves at three speeds, locate the best-efficiency point (BEP) at each speed, collapse the data with dimensionless coefficients, and classify the pump by its specific speed.

2. Theory

The pump head is the energy added per unit weight of liquid. Because the suction and delivery pipes have the same diameter, the velocity terms cancel and the head depends only on the gauge readings and the 0.18 m height between the gauges: $H_p = (p_d - p_s)/\gamma + (z_d - z_s)$. The useful hydraulic power is $P_h = \gamma Q H_p$, the brake (shaft) power is the motor electrical power times the motor efficiency, $P_b = 0.92\ P_{el}$, and the pump efficiency is $\eta = P_h/P_b$.

Table 1. Dimensionless groups used to compare speeds (n in rad/s, D = 0.104 m)

Quantity	Definition	Meaning
Head coefficient	$C_H = gH/(n^2D^2)$	Head scaled by impeller tip speed
Capacity coefficient	$C_Q = Q/(nD^3)$	Flow scaled by speed and size
Power coefficient	$C_P = P_b/(\rho n^3 D^5)$	Shaft power scaled by speed and size
Specific speed	$n_s = C_Q^{1/2}/C_H^{3/4}$	Impeller type at the BEP (dimensionless)
Specific speed (US)	$N_s = N \cdot Q^{1/2}/H^{3/4}$	rpm, gpm, ft; $N_s \approx 2733\ n_s$

When Reynolds-number effects are small, data taken at different speeds collapse onto single curves of C_H and C_P against C_Q. Radial-flow pumps have N_s of roughly 500–4000, mixed-flow pumps 4000–10,000, and axial-flow pumps above 10,000 (Munson et al., 2006).

3. Procedure

The pump was primed with the base-module pump, the speed was set with the potentiometer, and the delivery ball valve was opened fully. At each valve position the suction pressure, delivery pressure, and motor power were read once the gauges settled, and the flow was measured by timing a 30 s fill of the volumetric tank. The valve was then closed a step and the readings repeated until shut-off. Nine points were taken at 1800 rpm and seven each at 2500 and 3000 rpm, with the speed corrected between points as needed.



Figure 1. HM 150.04 trainer on the HM 150 base module



Figure 2. Pump, motor, delivery gauge and ball valve

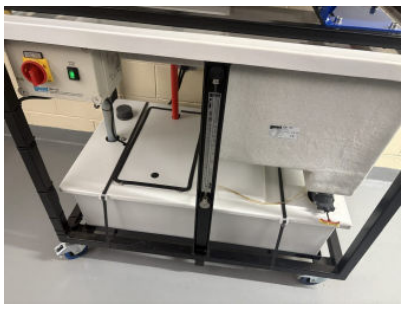


Figure 3. Volumetric measuring tank with sight gauge

4. Results

Characteristic curves

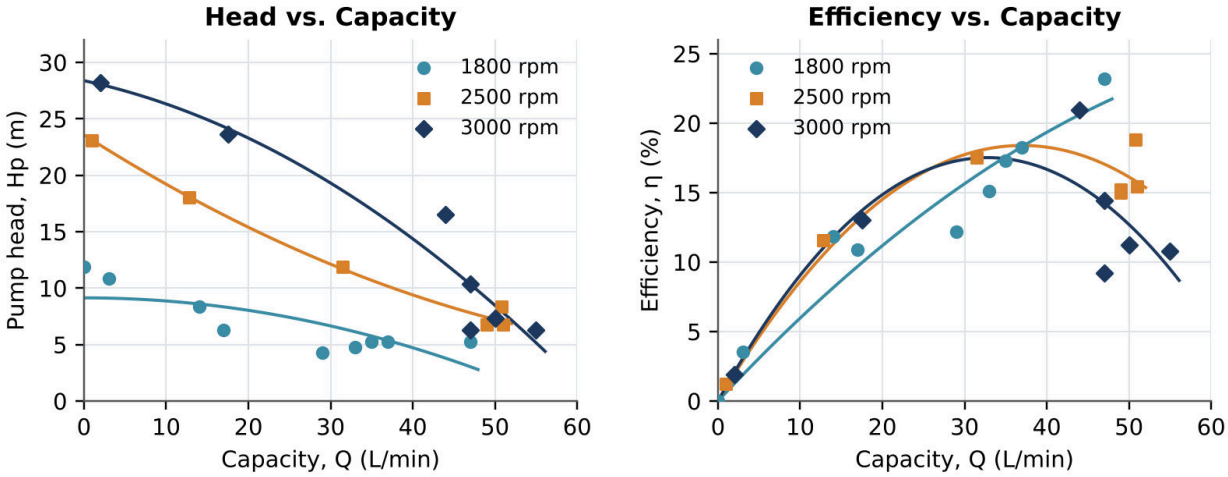


Figure 4. Pump head and efficiency vs. capacity at three speeds (points: measured; lines: quadratic fits, efficiency fits pass through the origin).

Table 2. Shut-off head, best-efficiency point, and specific speed at each speed

Speed (rpm)	Shut-off head (m)	Q at BEP (L/min)	H at BEP (m)	η_{max} (%)	n_s	N_s (US)
1800	11.9	47.0	5.28	23.2	0.273	747
2500	23.1	50.8	8.33	18.8	0.280	766
3000	28.2	44.0	16.49	21.0	0.188	513

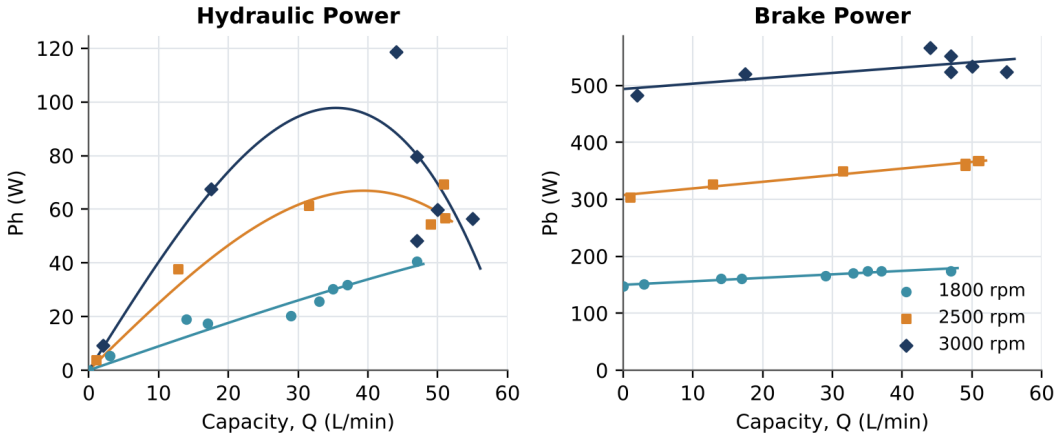


Figure 5. Hydraulic and brake power vs. capacity (lines: least-squares fits).

Dimensionless performance

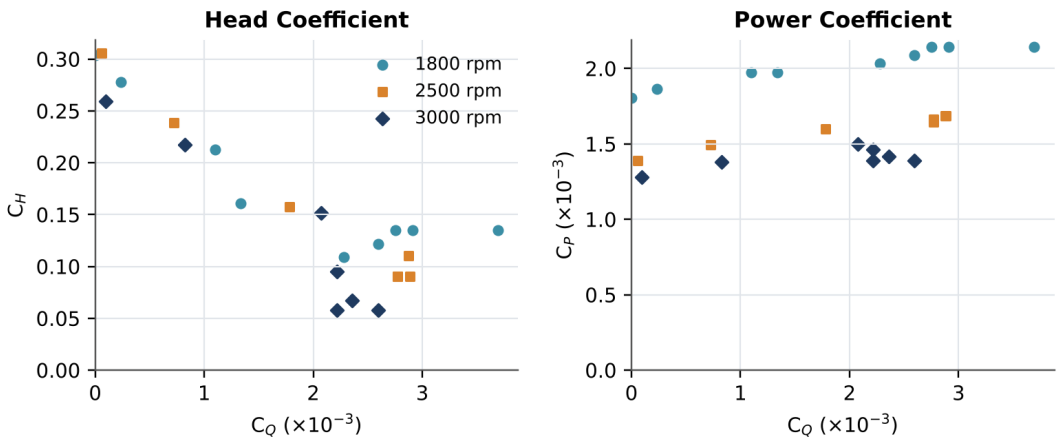


Figure 6. Head and power coefficients vs. capacity coefficient for all three speeds.

5. Discussion

Characteristic curves. The curves follow the expected shape for a centrifugal pump. At every speed the head falls steadily as flow increases, and raising the speed shifts the whole curve up and to the right. At 3000 rpm the head drops from 28.2 m at shut-off to 6.3 m at 55 L/min. Hydraulic power is zero at shut-off, peaks at 119 W near 44 L/min, and then falls, while brake power stays between about 483 and 566 W because internal losses are present even with no flow. As a result the efficiency rises to a single peak and then drops.

Efficiency and specific speed. The best efficiencies were low, about 19–21% at the two higher speeds, which agrees with the reference curve in the lab manual (about 20.8% at 2700 rpm). The specific speed at the BEP, $N_s \approx$ 513–766, sits at the bottom of the radial-flow range. Low-specific-speed, low-capacity pumps like this one are known to run at low efficiency (Pritchard & Mitchell, 2015). The 23.2% peak at 1800 rpm occurred with a delivery reading of 0 bar, so it is likely a gauge-resolution artifact rather than a real best point.

Dimensionless collapse and error sources. The power coefficients from the three speeds fall close to one band, and the head coefficients agree well at moderate and low flow, showing the affinity laws hold reasonably for this pump. The scatter near the fully open valve, and the shut-off head of 28.2 m against the manufacturer value of 36 m, come mainly from the class 1.6 Bourdon gauges, which are hard to read near zero, and from timing the tank fill by hand.

6. Conclusion

The HM 150.04 centrifugal pump was characterized at 1800, 2500, and 3000 rpm. The measured head, power, and efficiency curves match the theoretical behavior of a radial pump, and higher speed raised both head and capacity. At 3000 rpm the pump developed a 28.2 m shut-off head and peaked at 21% efficiency at 44 L/min and 16.5 m. A specific speed of $N_s \approx$ 513–766 classifies it as a low-specific-speed radial-flow pump, which is consistent with its low efficiency. Digital pressure transducers and an automated flow meter would reduce the scatter seen near the open-valve condition.

References

Moon, S. (n.d.). *Performance test of a centrifugal pump* [ME 405 laboratory manual]. New Jersey Institute of Technology.

Munson, B. R., Young, D. F., & Okiishi, T. H. (2006). *Fundamentals of fluid mechanics* (5th ed.). Wiley.

Pritchard, P. J., & Mitchell, J. W. (2015). *Fox and McDonald's introduction to fluid mechanics* (9th ed.). Wiley.